

gilmore-lie-groups: residuals

equations measured 2026-09-05T11:57:22Z against the model of 2026-09-03

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ E09584 (11.4) [conf 0.001]	188	■ 0.001 ● C +67	$\left.\left.\left.\left[\begin{array}{cccccccc} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & \ddots & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & -1 & \\ & & & & & & & 1 \\ & & & & & & & & -1 & \\ & & & & & & & & & \ddots \end{array}\right] \begin{array}{c} H_1 \\ H_2 \\ H_3 \\ \vdots \\ H_n \\ \frac{1}{\sqrt{2}}(E_\alpha + E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\alpha - E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_\beta + E_{-\beta}) \\ \frac{1}{\sqrt{2}}(E_\beta - E_{-\beta}) \\ \vdots \end{array}\right.\right.\right.$	(not rendered)	
gilmore-lie-groups_ E09778	243	■ 0.264 ●	$\begin{array}{l} \langle l \langle \theta L_- \theta' \phi' \rangle \\ \langle \theta L_- \theta' \phi' \rangle \\ \langle \theta \phi \rangle \end{array}$	$\langle \theta \phi \rangle L_- \theta' \phi' \rangle$ $\langle \theta \phi \rangle L_- \theta' \phi' \rangle$ $\langle \theta \phi \rangle L_- \theta' \phi' \rangle$	Matrix element of the angular momentum shift down operator in the coordinate representation: $\hbar(-\partial/\partial\theta + i(\cos\theta/\sin\theta)(\partial/\partial\phi))\delta(\cos\theta' - \cos\theta)\delta(\phi' - \phi)$. Matrix element of the angular momentum shift down operator in the algebraic representation: $\hbar\sqrt{(l' - m')(l' + m)}$ $\delta_{l'l}\delta_{m',m-1}$. Matrix element of the similarity transformation between the coordinate representation and algebraic representation. Also called spherical harmonic: $Y_m^l(\theta, \phi)$.
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

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gilmore-lie-groups_ E00381 (7.19)	119	■0.448 ●	$\begin{array}{c} \frac{i}{2} \left[\begin{array}{cc} 0 & ib_1 + b_2 \\ ib_1 - b_2 & 0 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} b_3 & 0 \\ 0 & -b_3 \end{array} \right] \\ \downarrow \text{EXP} \quad \downarrow \text{EXP} \\ \left[\begin{array}{cc} e^{+ib_3/2} & 0 \\ 0 & e^{-ib_3/2} \end{array} \right] \end{array}$	$\frac{i}{2} \left[\begin{array}{cc} 0 & ib_1 + b_2 \\ ib_1 - b_2 & 0 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} b_3 & 0 \\ 0 & -b_3 \end{array} \right]$ EXP ↓ ↓ ↓ EXP $[SU(1, 1)/U(1)] \times \left[\begin{array}{cc} e^{+ib_3/2} & 0 \\ 0 & e^{-ib_3/2} \end{array} \right]$	
gilmore-lie-groups_ E00590 (11.11)	190	■0.454 ●C -38	$\begin{array}{c} \frac{1}{2} \left[\begin{array}{cc} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{array} \right] = \frac{1}{2} \left[\begin{array}{cc} a_3 & 0 \\ 0 & -a_3 \end{array} \right] + \frac{1}{2} \left[\begin{array}{cc} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{array} \right] \end{array}$	$\frac{i}{2} \left[\begin{array}{cc} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{array} \right] = \frac{i}{2} \left[\begin{array}{cc} a_3 & 0 \\ 0 & -a_3 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{array} \right]$	
gilmore-lie-groups_ E00369 (7.7)	115	■0.468 ●	$\begin{array}{c} \frac{1}{2} \left[\begin{array}{cc} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{array} \right] = \frac{1}{2} \left[\begin{array}{cc} a_3 & 0 \\ 0 & -a_3 \end{array} \right] + \frac{1}{2} \left[\begin{array}{cc} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{array} \right] \end{array}$	$\frac{i}{2} \left[\begin{array}{cc} a_3 & a_1 - ia_2 \\ a_1 + ia_2 & -a_3 \end{array} \right] = \frac{i}{2} \left[\begin{array}{cc} a_3 & 0 \\ 0 & -a_3 \end{array} \right] + \frac{i}{2} \left[\begin{array}{cc} 0 & a_1 - ia_2 \\ a_1 + ia_2 & 0 \end{array} \right]$	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ EQ0006 (1.3)	18	0.944	$\begin{array}{l} \begin{array}{r} I \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \\ \quad (123) \rightarrow \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array}\right] \\ \quad (321) \rightarrow \left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array}\right] \\ \quad (12) \rightarrow \left[\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right] \\ \quad (23) \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right] \\ \quad (13) \rightarrow \left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array}\right] \end{array} \\ \begin{array}{l} \begin{array}{r} I \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \\ \quad (123) \rightarrow \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array}\right] \\ \quad (321) \rightarrow \left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array}\right] \\ \quad (12) \rightarrow \left[\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right] \\ \quad (23) \rightarrow \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right] \\ \quad (13) \rightarrow \left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array}\right] \end{array} \end{array}$	$\begin{array}{l} I \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (123) \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (321) \rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \\ (12) \rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (23) \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (13) \rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \end{array}$	
gilmore-lie-groups_ EQ0113 (3.3)	49	0.991 C -5	$\begin{array}{l} \begin{array}{cc} c=(a,b)=a+ib & a \in \mathbb{R}, b \in \mathbb{R}, i^2=-1 \\ (a,b) \longrightarrow \begin{bmatrix} a & b \\ -b & a \end{bmatrix} & i=(0,1) \longrightarrow \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \end{array} \\ \begin{array}{l} \begin{array}{cc} c=(a,b)=a+ib & a \in \mathbb{R}, b \in \mathbb{R}, i^2=-1 \\ (a,b) \longrightarrow \begin{bmatrix} a & b \\ -b & a \end{bmatrix} & i=(0,1) \longrightarrow \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \end{array} \end{array}$	$\begin{array}{l} c=(a,b)=a+ib \quad a \in \mathbb{R}, \ b \in \mathbb{R}, \ i^2=-1 \\ (a,b) \longrightarrow \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \quad i=(0,1) \longrightarrow \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \end{array}$	
Conf. MathPix confidence: Residual render vs scan (inkdrill):					

Low confidence

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ EQ0249 [conf 0.031]	84	■ 0.031 ●	\begin{aligned} f_{\{0\}}(\backslash\theta) &=\backslash\cos \backslash\theta & \backslash\text{\\ or } & f_{\{0\}}(\backslash\theta) \end{aligned}=\backslash\cos \backslash\theta \backslash\text{\\ } \{ \} \{ \} ^{\backslash\left(f_{\{1\}}(\backslash\theta)\backslash\right.}=\backslash\sin (\backslash\theta) / \backslash\theta \quad \backslash\begin{aligned} & \backslash\theta f_{\{1\}}(\backslash\theta)= \backslash\sin (\backslash\theta) \backslash\backslash f_{\{2\}}(\backslash\theta) & =(1-\backslash\cos (\backslash\theta)) / \backslash\theta^{\{2\}} \end{aligned} \quad \backslash\theta^{\{2\}} f_{\{2\}}(\backslash\theta)=1-\backslash\cos (\backslash\theta),	$f_0(\theta) = \cos \theta$ or $f_0(\theta) = \cos \theta$ $f_1(\theta) = \sin(\theta)/\theta$ or $\theta f_1(\theta) = \sin(\theta)$ $f_2(\theta) = (1 - \cos(\theta))/\theta^2$ or $\theta^2 f_2(\theta) = 1 - \cos(\theta)$	$f_0(\theta) = \cos \theta$ or $f_0(\theta) = \cos \theta$ $f_1(\theta) = \sin(\theta)/\theta$ or $\theta f_1(\theta) = \sin(\theta)$ $f_2(\theta) = (1 - \cos(\theta))/\theta^2$ or $\theta^2 f_2(\theta) = 1 - \cos(\theta)$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					